Current MRI biomarker validation

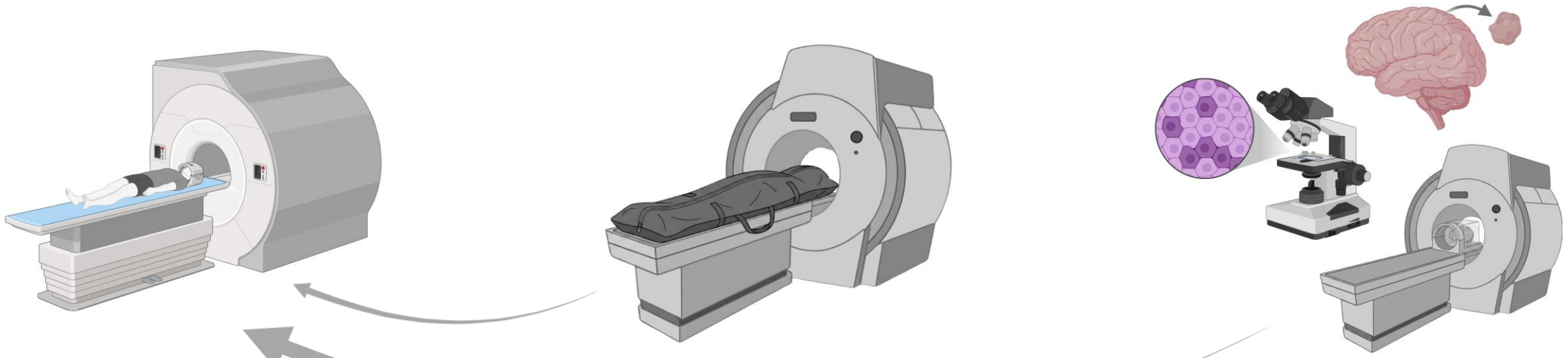
Living

Post mortem

In vivo data collection

In situ data collection
(in our case)

Ex situ data collection
(Formalin fixed / rehydrated)

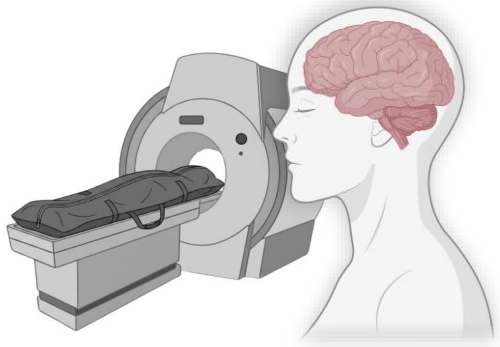


Validation of in vivo MRI biomarkers

Do formalin-fixed and rehydrated tissues still represent the in vivo tissue?

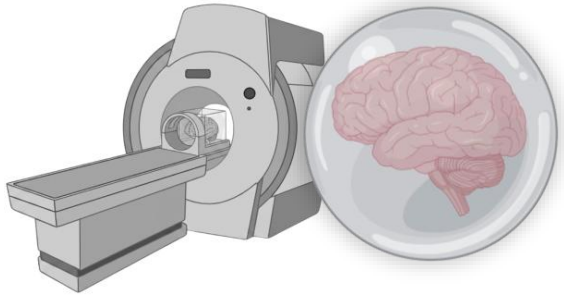
Methods

In situ data collection



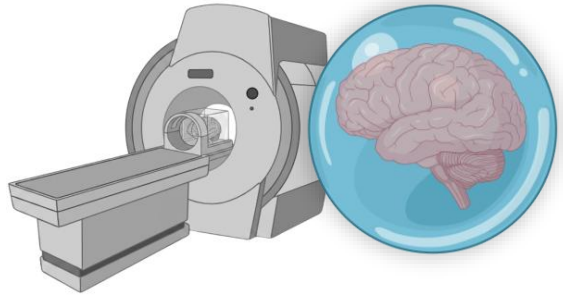
Formalin fixation
(4,5 %, 3 months)

Ex situ data collection
(Formalin fixed)



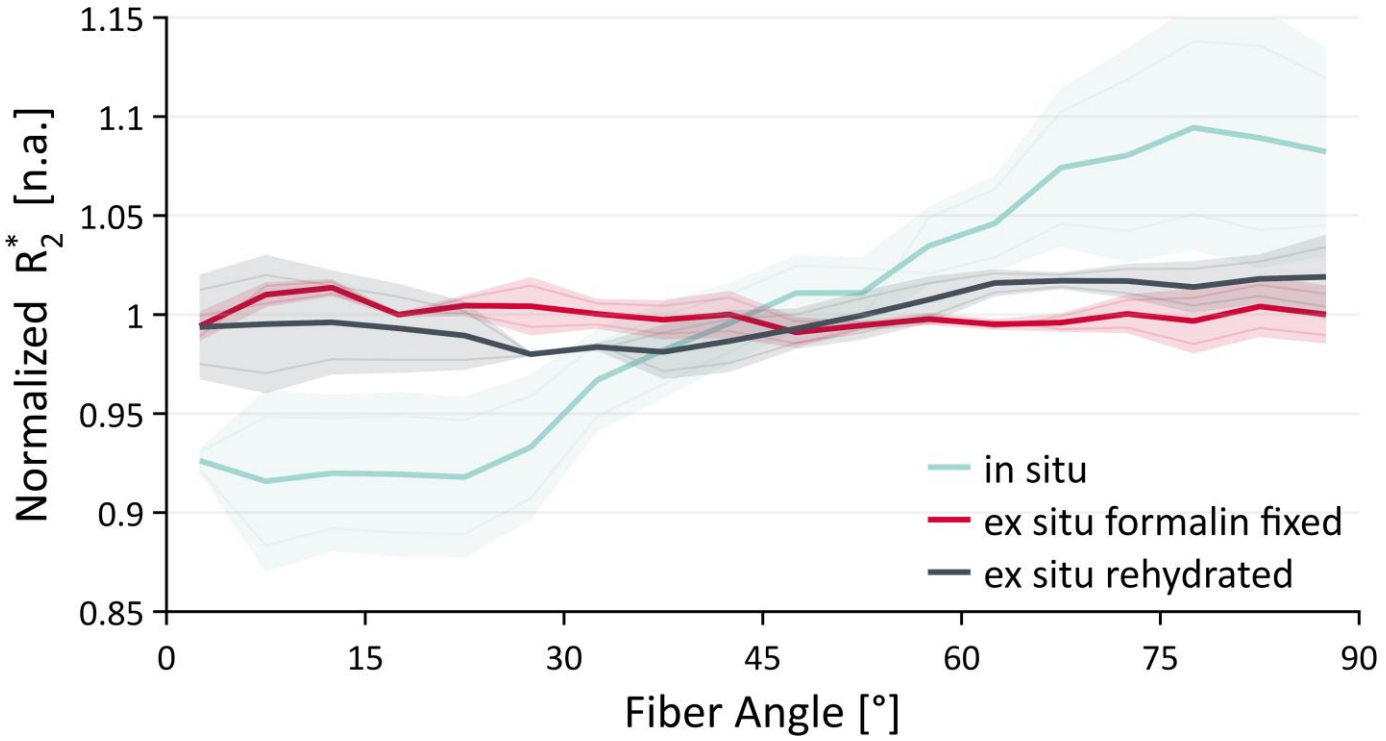
PBS rehydration
(9 days, 3 fluid changes)

Ex situ data collection
(Rehydrated)



3T MRI Sequences: MP2RAGE (structural imaging, segmentation), ME-GRE (R_2^* mapping), Multi-shell DTI (fiber orientation)

Fixation effects are only partially reversible



Formalin fixation

- Fractional anisotropy ↓
- Mean-, axial- & radial-diffusivity ↓
- R_2^* ↑

Rehydration

- Fractional anisotropy ↓
- Mean-, axial- & radial-diffusivity ↑
- R_2^* ↓

→ **Fixation and rehydration limit translation to in vivo conditions**

Interested in the full story?
Full results at poster 19